Blindfold

Run AI on computers you don’t trust. They never see your input, and they get caught if they cheat.

Stays on your computer	Goes to each server
Your actual text	A random-looking piece of it
The final answer	The public model + random masks

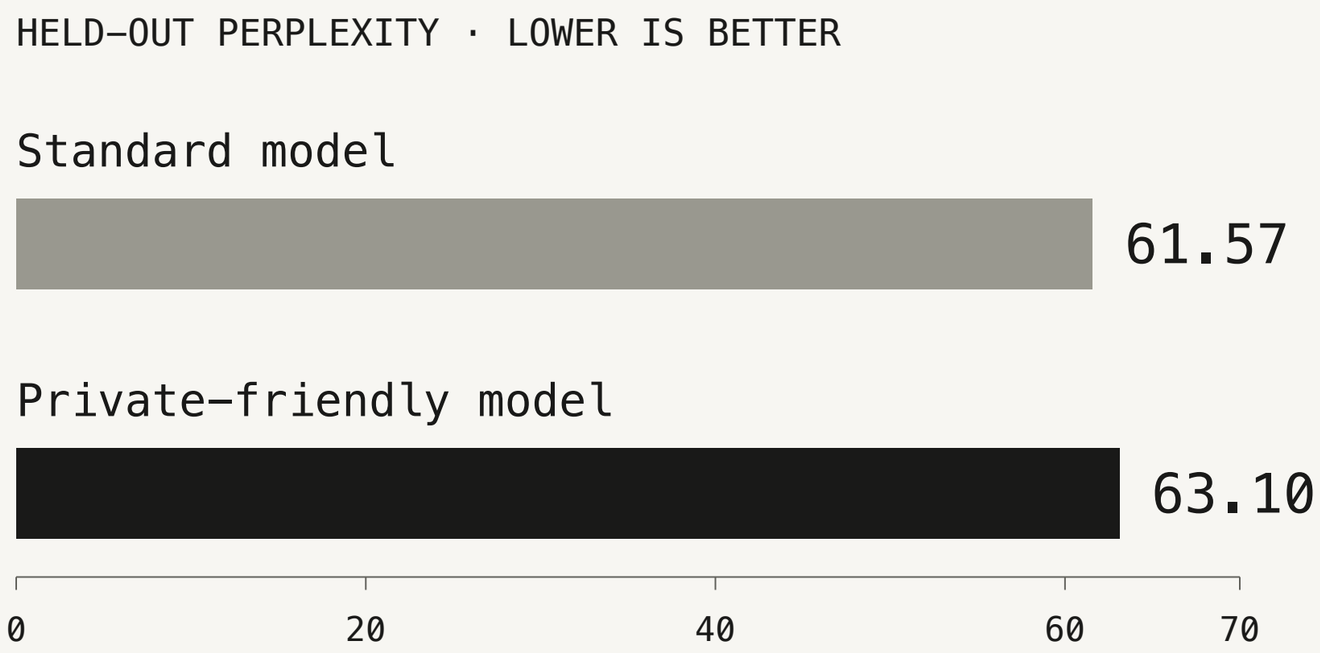
Who can read your input

If these servers' pieces are combined	Can your input be read?
Any one server	No
All but one of the servers	No
All of the servers	Yes

A server that changes one number is caught before its answer is used.

Assumes the servers are run by different people

Real AI models can be rebuilt for this at only 2.48% quality cost



6,164,096

parameters in each test model

14,336,000

training tokens

Rebuilt model uses only add, multiply and square.

Measured: perplexity on held-out text · data/quality.json

Separate model from the demo

The model only adds, multiplies and squares

Step	Who does it	Messages
Split the input into pieces	Your computer	Pieces go to the servers
Multiply by public weights	Each server, on its own	None needed
Square	Servers, through your computer	Checked before it’s used
Multiply again, then check	Servers, then your computer	Answer pieces come back
Per character		2 round trips

From engine/device.py and engine/node.py

Built today, and where it goes

Built today	Next
A 19,295-parameter note model	Real-size models on rented GPUs
Any number of servers (tested 2–7), over the network or USB	Servers run by different people
Same output as the normal model; cheating caught live	A full security review
Your computer makes the random masks	Servers make them and talk directly